

2026 FIFA World Cup Predictor Project Report

Chase Foster (chase_foster@berkeley.edu), Leah Finger (leah_finger@berkeley.edu)

Nicole Yokota (nyokota@berkeley.edu), Kevin Rho (kevinrho51@berkeley.edu)

Jay Yao (jayyao@berkeley.edu)

1.0 Abstract

Arguably the world's biggest sports festival, the FIFA World Cup, starts in June 2026 and will be held in the United States, Canada, and Mexico. As we approach the event, we believe it will be a fantastic opportunity to apply what we learned in this course to build ML models to predict the outcomes of World Cup games. Although the World Cup starts after this course ends, we look forward to directly assessing our model's performance this June.

This project builds a pipeline to predict match outcomes: draw, home win, or away win using historical FIFA match results from 1992 to 2022 combined with world rankings, custom ELO ratings, and dynamic form features. We implemented and compared these models: baseline, logistic regression, a multi-layer neural network, and tree-based models (e.g., random forest, gradient boost). All of our advanced models performed similarly, with accuracies plateauing at 61%, but given that the tree-based model is an industry standard for sports and additionally achieved the highest draw recall, we have selected this model as our final model.

2.0 Introduction

Forecasting the outcome of soccer matches between nations has significant interest for sport analytics, betting markets, and tournament planning. However, the task is difficult as soccer is low scoring, commonly results in draws, teams fluctuate in quality, outcomes can often be prone to influences that are difficult to model (e.g. effects of home team advantages can vary by competition tier), and countries don't play against one another as often as club teams play against one another (such as in the Premier League).

We frame this as a supervised multi-class classification task. The input is a fixed-length feature vector representing a single match between two teams, encoding team quality signals (ELO ratings, FIFA rankings), head-to-head game results, and recent form (goals per game). The output will be: 0 = Draw, 1 = Home team wins, 2 = Away team wins. The dataset spans 30 years of matches from 1992 to 2024, giving us 25,691 labeled examples after filtering for matches where FIFA rankings exist for both teams. The same feature engineering and modeling approach can be used to generate pre-match win probabilities for any upcoming fixture between two countries for which we have the data. With the 2026 World Cup starting in June, pre-match predictions have direct real-world stakes.

3.0 Related Work

There is an extensive body of work applying machine learning to sports match outcome prediction, with substantial variation in feature design, modeling algorithms, and classification type. A key distinction across this literature concerns the timing of feature availability: some studies rely on match performance indicators that are only observable after a match, while others constrain features to pre-match information so that models can predict upcoming matches.

Song et al. (2024) exemplify a post-match approach, using 14 match performance indicators and evaluating multiple algorithms, including Decision Trees, Logistic Regression, AdaBoost, Random Forests, and an Artificial Neural Network (ANN). They formulate the task as a binary classification problem ("winning" vs. "non-winning") and report that the ANN provides the strongest predictive performance. A major strength of this study is its use of SHAP analysis, where the authors identify "On Target," "Shooting Opportunity," and "Ball Progressions" as the most influential predictors of outcomes. This analysis improves model interpretability and also provides actionable insight by highlighting technical dimensions that coaches can prioritize in training. However, these indicators are not available prior to kickoff and are therefore not applicable to our goal of predicting upcoming match outcomes. In addition, the model is trained on 59 matches from the 2022 World Cup, which yields a focused tournament context but may also reduce stability and generalizability due to the small sample. Finally, because draws are valid outcomes in the World Cup group stage, our work extends beyond this study by training a three-class model (Draw / Home win / Away win).

In contrast, Ramadhani et al. (2026) adopt a pre-match modeling framework using international match data from 1993-2022. They engineer features intended to be available prior to matches, including contextual features, FIFA ranking and points differences, and

four “squad strength” difference variables (goalkeeper, defense, midfield, and offense). They evaluate Logistic Regression, Random Forest, and HistGradientBoostingClassifier (HistGBM) with a 3-class target variable (‘win’, ‘draw’ or ‘loss’ for the home team) selecting HistGBM as their final model. Although their model identifies home-team wins well (win recall of 78%), it performs poorly on draw outcomes (draw recall of 0.08), illustrating the persistent difficulty of draw prediction in soccer. A notable strength of this study is its attempt to incorporate player-adjacent information via squad strength metrics, which may capture variation not fully reflected by rankings alone. However, it is unclear how these squad strength scores are computed, and such data may not be consistently available across teams or time periods, which may limit reproducibility.

Building on the insight that relative-strength features are effective, we incorporate both FIFA ranking and points differences, and additionally, sums. However, rather than relying on potentially unavailable squad ratings, we introduce dynamic form features constructed from each team’s previous five matches. These features are designed to capture short-term changes in team performance prior to a match and to complement FIFA rankings, which may be insensitive to recent shifts. Motivated by the draw-prediction challenges observed in Ramadhani et al. (2026), we evaluate whether incorporating recent form features improves performance for draws while acknowledging that draw outcomes may remain intrinsically difficult to predict due to the low-scoring and high-variance nature of soccer matches.

4.0 Data

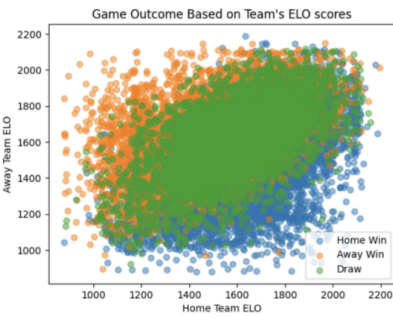


Figure 1: Game Outcomes based on the Team’s ELO scores

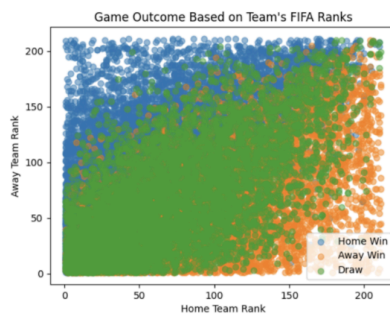


Figure 2: Game Outcomes based on the Team’s FIFA Rankings

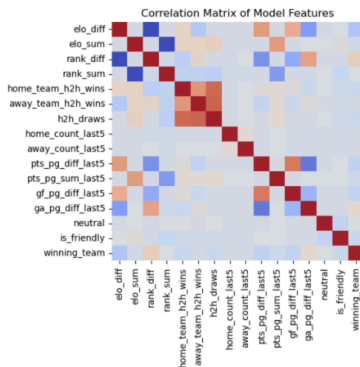


Figure 3: Correlation Matrix of Model Features

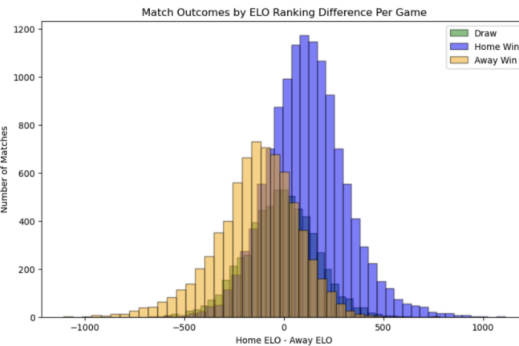


Figure 4: Histogram of Match Outcomes by ELO Ranking Difference Per Game

Our primary dataset from Kaggle includes 49,071 international soccer matches (1872-2026, 9 features), which we supplement with the FIFA rankings dataset (1992-2024, 13,130 records, 9 features). Because rankings are central to our model, we restrict training to matches from 1992 to 2024, while computing all historical features beforehand to preserve team performance context. After merging, filtering for matches with available rankings for both teams, and completing feature engineering, the final dataset contains 25,691 records with 15 features.

Our target variable is match outcome (draw, home win, away win), and the inputs capture team strength, recent form, and match context. Key features include the difference and sum of ELO ratings (dynamically updated after each match based on opponent strength, home advantage, match type, goal difference, and result), as well as the difference and sum of FIFA rankings, where the sum provides additional context on overall match quality.

To ensure only pre-match information is used, we also derive features from each team’s previous five games, including normalized points per game (3/1/0 system) and total goals scored and conceded. Two binary features further indicate whether the match was played at a neutral venue and whether it was a friendly or competitive match.

To better reflect real-world forecasting, we adopt a time-based split rather than a random split: matches before September 7, 2012 form the training set, those between September 7, 2012 and January 12, 2019 form the validation set, and matches after January 12, 2019 make up the test set, resulting in a 60:20:20 split (15,404 / 5,147 / 5,140).

Figures 1 and 2 show a clear pattern: teams with stronger ELO ratings or higher FIFA rankings are more likely to win, while draws tend to occur between evenly matched teams. Given how closely these features track one another, we note potential collinearity. This is supported by Figure 3, where the correlation matrix identifies rank_diff as the most predictive feature, and further reinforced by Figure 4, which shows that winning teams generally have higher ELO ratings, with draws distributed around near zero difference.

5.0 Methods

To predict 2026 FIFA World Cup match outcomes, we implemented a baseline model and three machine learning models: a logistic regression model, a neural network, and a decision tree. These models were selected to balance interpretability, performance, and the ability to capture nonlinear relationships in structured match data.

Our baseline model predicts match outcomes (draw / home win / away win) by selecting the team with the higher ELO score as the winner. This approach serves as a simple but meaningful benchmark, as ELO rankings are widely used to approximate team strength. If our more complex models fail to outperform this baseline, it would suggest that additional engineered features do not provide significant predictive value.

The first machine learning model is a multinomial logistic regression model implemented in TensorFlow. This model predicts the probability of each outcome class using a linear combination of input features followed by a softmax activation function. Logistic regression provides interpretability by allowing us to understand the influence of input features, such as ELO difference, on the outcome. It also serves as a strong starting point for comparing more complex nonlinear models.

The second machine learning model we implemented is a feedforward neural network (FFNN). This model consists of multiple fully connected layers that transform input features through nonlinear activation functions and learned weights. FFNN was selected because it can capture nonlinear interactions between features, and it provides greater expressive power than logistic regression. FFNN also supports interpretability through the learned weights, which can provide insight into the relative importance of each feature.

The final machine learning model we implemented is a decision tree. For this model, we also explored different methods such as random forests and Gradient Boosting. Decision trees recursively partition the feature space by selecting splits that maximize information gain. The different methods, such as random forests and Gradient Boosting, improve this by combining multiple decision trees. Specifically, random forests reduce variance through averaging, and Gradient Boosting improves performance by sequentially correcting errors. A tree-based model was selected because they naturally handle nonlinear relationships and feature interactions, are robust to outliers and noisy data, and are great to work with smaller-sized tabular datasets, complementing our project setup.

6.0 Experiments

6.1 Baseline & Logistic Regression

We began by building a rule-based baseline model: predict the team with the higher ELO rank to win, the lower rank to lose, and equal ranks to draw. The accuracy of our baseline model on the training dataset is 55%, serving as a benchmark for evaluating our models. For the logistic regression model, we used a softmax activation function for multiclass classification, and implemented the Keras Tuner to find the optimal learning rate. This model saw improved performance over the baseline model with a final training accuracy of 60.25% over 30 epochs, and a test accuracy of 60.89%, demonstrating strong generalizability of this model on unseen data.

6.2 Feedforward Neural Network Model

The FFNN was implemented in TensorFlow/Keras as a sequential multilayer perceptron with three fully connected hidden layers (256, 128, and 64 units) using ReLU activations, followed by a 3-unit softmax output layer to produce class probabilities. Training used sparse categorical cross-entropy loss and stochastic gradient descent (SGD; learning rate = 0.002) over 20 epochs with a batch size of 64. Performance was consistent across all splits, achieving an accuracy of 60.26% on the training set, 60.62% on the validation set, and 60.82% on the test set, indicating limited overfitting and strong generalization capabilities. Despite extensive hyperparameter tuning, including learning rates, epoch counts, batch sizes, hidden-layer depths and widths, batch normalization, dropout, alternative

optimizers and even class weights (to increase draw predictions), the best-performing model remained this comparatively simple architecture, with predictive performance plateauing at 61% accuracy.

6.3 Tree-Based Model

In an effort to capture nonlinear relationships between our features and the outcome, we decided to explore tree-based models. To ensure that our models attempted to fairly predict the ‘draw’ and ‘away win’ classes, we implemented specific class weights for the outcome classes. The ‘draw’ and ‘away win’ classes were assigned a 1.5 weight, and the ‘home win’ class was assigned a 1.0 weight. We began with a Random Forest Classifier with `n_estimators = 500` and `max_depth = 7`. This model resulted in a training accuracy of approximately 63.33% and a validation accuracy of 59.96%. We then implemented a Gradient Boosting Classifier with `n_estimators = 300`, `max_depth = 3`, and `learning_rate = 0.05`. This model resulted in a training accuracy of approximately 62.95% and a validation accuracy of approximately 60.83%. Both models appear to generalize well due to the small differences between the training and validation accuracies; however, neither model seemed to significantly outperform the baseline model. Upon examination of the confusion matrices for these two models, we found that both struggled to accurately predict the draw class. We attempted many different configurations of the model parameters; however, we were not able to achieve a validation accuracy greater than 61%, and each configuration continued to struggle with accurately predicting the draw class. Both models performed similarly, but the Gradient Boosting model had a slightly higher overall accuracy and it was better at predicting the ‘home win’ class, so we selected it for the final tree-based model, achieving a 60.58% test accuracy. Since our models struggled to predict the draw class, we also implemented a XGBoost Binary Classifier, which was tasked with predicting one of two outcomes: ‘home win’ or ‘no home win’. The ‘no home win’ class included both the ‘draw’ and ‘away win’ outcome classes. This model resulted in a 78.84% training accuracy and a 72.29% validation accuracy. This notably improved performance is almost certainly due to combining the ‘draw’ and ‘away win’ classes, as the model no longer had to attempt to predict a ‘draw’ outcome, which has proven to be a major challenge. For consistency, this model was only used to gain insights and was not selected for our final tree-based model despite its performance.

6.4 Comparisons

Model	Training Accuracy	Validation Accuracy	Test Accuracy	Draw Recall
Baseline	55.15	55.76	59.44	0
Logistic Regression	60.25	60.79	60.89	0.006
Neural Network	60.26	60.62	60.82	0.025
Tree-Based (Gradient Boost)	62.95	60.83	60.58	0.077

Table 1: Summary of performance metrics from all models

All models remained consistent in performance and demonstrated some improvement over the baseline model. The baseline model achieved a test accuracy of 59.44%, but this is likely due to a class imbalance of more ‘home win’ outcomes. All of the models additionally improved test accuracy, but only by a narrow margin. We include draw recall in Table 1 to illustrate how each model severely struggles to identify draw outcomes, but also highlight that the Gradient Boosting model performs best on this metric. The overall small performance differences across all of the models highlight the difficulty of predicting World Cup match outcomes.

7.0 Conclusions

All of our models showed strong generalization capabilities and essentially no evidence of overfitting, as each model performed just as well on unseen data. The use of tree-based models is considered an industry standard for predicting the outcome of sports games because tree-based models are effective at detecting nonlinear patterns in the data, selecting features that are most important for predicting the game outcome, and splitting across those key features. For this reason, in addition to achieving the highest draw recall, we would proceed with deploying the Gradient Boosting model. Simultaneously, we acknowledge that we ultimately could choose any of the three models and all would perform similarly, given that no model was able to surpass 61% overall accuracy when predicting

the three output classes. Based on each model's low draw precision and in particular, very low draw recall scores, accurately predicting a 'draw' outcome has demonstrated to be a formidable task for our models, and is evidently the main culprit behind this apparent ceiling on accuracy. If we instead formulated a binary task of predicting whether or not the home team wins, we expect all of our models to show significant improvements in performance, as demonstrated with the XGBoost Binary Classifier. However, since draws are allowed in the group stage of the World Cup, we wanted to build models that included predictions for draws, and we recognize that our results suggest that our features may not sufficiently support this complex task.

Other limitations in this study may include our lack of data on many of the intricate variables that make up a team's strength, such as player information, player injuries, goal-scoring abilities, penalties and penalty shots, team ball possession time, and other variables that may be important in determining the winner of a match. Additionally, with our goal of predicting upcoming matches, we will always face the natural limitation of being crucially constrained to information available prior to kick-off. It is also difficult for our models to predict outcomes that defy natural patterns and structures, such as the occasional 'upset', which is when the unexpected team, or 'underdog', wins. To improve accuracy scores on the three-outcome class models, more complex and nuanced data may be necessary, such as specific player-level data relevant to the time of a given match. We made efforts to increase model robustness by calculating custom ELO rankings and engineering dynamic form features, but further enriching our models with additional datasets that can support more sophisticated feature engineering may promote improved performance in future research.

8.0 Contributions

Chase Foster: Researched ML models, pair-programming, tree-based models, Methods, Experiments, & Model Comparison sections.
Leah Finger: Feature selection, EDA, Data Preprocessing, Visualization, pair-programming, baseline and logistic regression models, conclusion.

Nicole Yokota: Feature selection, EDA, Data Preprocessing, Visualization, pair-programming, neural networks, related works section.

Kevin (Shinkyum) Rho: Feature selection, EDA, Data Preprocessing, repo set up, pair-programming, neural networks, data section writeups.

Jay Yao: Abstract, Introduction sections, pair programming, tree-based models, introduction writeups.

Notebook Ownership: We've pair-programmed all together and therefore have only the common notebooks in our repository.

9.0 Citations

Ramadhani, Z. I., Syaifudin, Sahfitda, B., Kusuma, S., & Ardiyansyah. (2026). Application of Machine Learning in Predicting FIFA World Cup Matches. *Journal of Artificial Intelligence and Engineering Applications*, 5(2).

<https://ioinformatic.org/index.php/JAIEA/article/download/1918/1327>

Song, Y., Sun, G., Wu, C., Pang, B., Zhao, W., & Zhou, R. (2024). Construction of 2022 Qatar World Cup match result prediction model and analysis of performance indicators. *Frontiers in Sports and Active Living*, 6, 1410632.

<https://doi.org/10.3389/fspor.2024.1410632>

10.0 Repository Link

Github Repo Link: <https://github.com/kevinrho51/207-Spring-World-Cup-Predictor/tree/main>

11.0 Misc.

The figures below are our model's predictions for the 2026 FIFA World Cup group-stage winners and the knockout stage, based on the real-world cup schedule and matchups. Please refer to our World Cup prediction notebook for how we generated our predictions: https://github.com/kevinrho51/207-Spring-World-Cup-Predictor/blob/main/src/wc_2026_predictions.ipynb

GROUP A 1 MEX 9 pts 2 KOR 6 pts	GROUP B 1 CAN 9 pts 2 SUI 6 pts	GROUP C 1 BRA 9 pts 2 MAR 6 pts	GROUP D 1 USA 9 pts 2 AUS 6 pts
GROUP E 1 GER 9 pts 2 ECU 6 pts	GROUP F 1 NED 9 pts 2 JPN 6 pts	GROUP G 1 BEL 9 pts 2 IRN 6 pts	GROUP H 1 ESP 9 pts 2 URU 6 pts
GROUP I 1 FRA 9 pts 2 SEN 6 pts	GROUP J 1 ARG 9 pts 2 ALG 6 pts	GROUP K 1 POR 9 pts 2 COL 6 pts	GROUP L 1 ENG 9 pts 2 CRO 6 pts

Figure A: Group Stage Outcomes

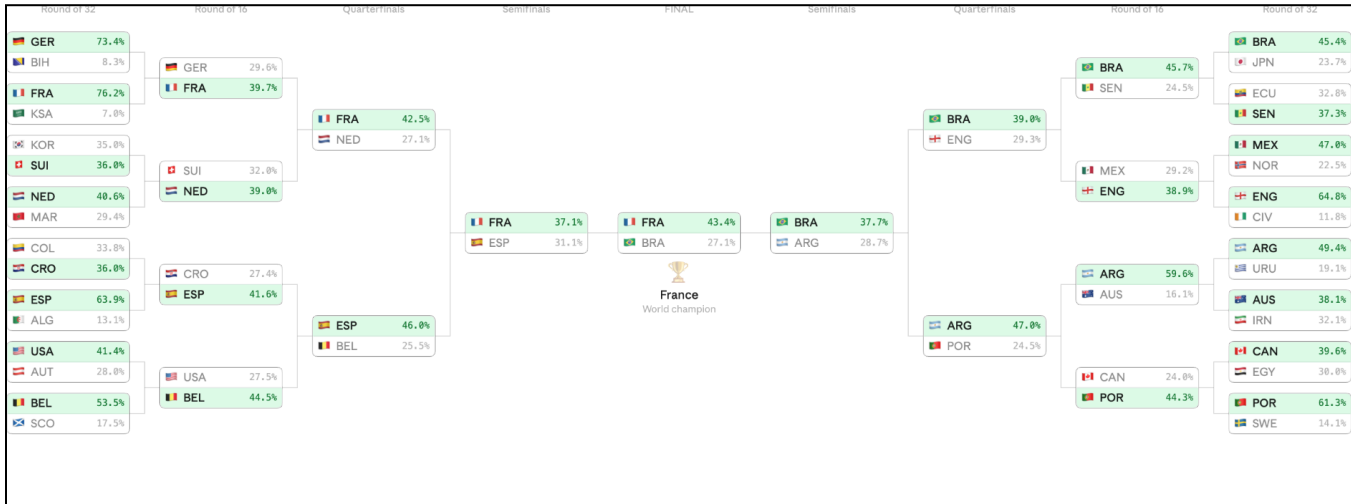


Figure B: Knockout Stage Predictions